

You should also go through the available configuration options for your laptop's wireless card, as some of them give you the option of increasing the transmission power of the card. Remember that certain laptop models reduce the power sent to peripherals when running in their battery savings mode, the best way to tell is by noticing whether the range seems to improve when you start charging the laptop.

Can I use Wi-Fi to join existing wired LANs?

Yes, you can use multiple access points running in bridging mode to bridge connections between various wired LANs. However, since wireless access points run a lot slower than their wired counterparts, the access speeds from one LAN to another might not be up to the mark. Unless using a hardware switch or router is forbidden or impossible, you should not attempt to bridge two wired LAN networks using this solution.

Why does everyone recommend Wi-Fi, and then warn you about the possible security flaws?

Wireless has given us a solution to problems that could not be fixed earlier. Wireless can also save a lot of money, by doing away with cabling costs, and the need to fill out a hundred forms, while standing in a never-ending queue inside a government office, in order to get permission to dig up a road to lay your cables. It's the same reason most of us use the Windows operating system; we are all more at threat when using Windows than any other OS, but its ease of use and simplicity has us hooked.

What is ad-hoc mode, and is it advisable to use it?

Devices running in ad-hoc mode can connect to each other directly, without needing to connect to an access point. All devices running in ad-hoc mode can discover each other and communicate with each other to form an unorganised network. The limitation here is that all the devices must share the same SSID and operate on the same Wi-Fi channel. The disadvantages of such networks is that the performance takes a hit as the number of devices increases, and there is no way to properly manage, and thus protect, the users of such a network.